



11687 - EXHALE: Expelling the X-ray High-energy winds to InfrAred-traced gaLactic scale fEedback - Tracing multiphase outflows in Mrk 877

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	Mrk877-NIRSpec	NIRSpec IFU Spectroscopy	(3) Mrk877-NIRSPEC
	2	Mrk877-MIRI	MIRI Medium Resolution Spectroscopy	(1) Mrk877-MIRI
	3	Mrk877-MIRI-BACKG ROUND	MIRI Medium Resolution Spectroscopy	(2) Mrk877-MIRI-BACKGROUND

ABSTRACT

EXHALE (Expelling the X-ray High-energy winds to InfrAred-traced gaLactic scale fEedback) is a pilot JWST program designed to trace how X-ray-detected Ultra-Fast Outflows (UFOs) couple to the host interstellar medium and regulate star formation. The nearby quasar Mrk 877 ($z = 0.11$) hosts a confirmed UFO detected by XMM-Newton, making it an ideal step to study how accretion-driven winds evolve into large-scale feedback.

JWST Proposal 11687 (Created: Thursday, June 11, 2026, 1:56:59PM Eastern Standard Time) - Overview

Using JWST/NIRSpec IFU and MIRI MRS, EXHALE will map the kinematics and excitation of the ionized, coronal, and molecular gas outflows through key infrared tracers on sub-kpc scales. Mrk 877 bridges the gap between most luminous quasars like PDS 456 and more typical Seyferts like NGC 4151, offering a unique test of AGN feedback efficiency. By combining X-ray UFO energetics with JWST-resolved outflow diagnostics, EXHALE will provide one of a few views of multi-phase wind coupling and establish a methodological benchmark for a future JWST survey connecting accretion-driven winds to star-formation quenching in galaxies.

OBSERVING DESCRIPTION

We will observe Mrk 877 using the NIRSpec and MIRI IFUs with the following settings:

-NIRSpec IFU with G235H/F170LP

Target acquisition starts on the quasar itself with the WATA/F140X filter

9 Cycle dither pattern

15 groups and 3 integrations per dither position

Readout mode: NRSRAPID

--> total exposure time on source: 4638.28 sec

1 leakage exposure at 1 dither position: 515.36 sec

-MIRI IFU with the SHORT, MEDIUM, and LONG wavelength settings

4-point all-wavelength extended source dither pattern

35 groups and 7 integrations per dither position for each wavelength setting

Readout mode: FAST

--> total exposure time on source: $3 \times 2786.145 \text{ s} = 8358.42 \text{ sec}$

1 off-source background data cube

(obtained immediately after the on-source cube using the same settings as the on-source data cube)

--> total exposure time on background: $3 \times 2786.145 \text{ s} = 8358.42 \text{ sec}$

--> exposure time for simultaneous interspaced F770W images: 85.46 sec each

Updates after IS review:

- NIRSpec observation:

Canceled the WATA.

JWST Proposal 11687 (Created: Thursday, June 11, 2026, 1:56:59PM Eastern Standard Time) - Overview

Updated the readout pattern to NRSIRS2RAPID with 11 groups, 3 integrations, and 9 dithers.

Added PA restrictions of 23–111 deg, 139–140 deg, 233–244 deg, and 338–349 deg.

- MIRI observation:

Reordered the MIRI observations so that the background observation is taken before the science observation.

Updated the MIRI MRS detector setup to 42 groups and 6 integrations.

Updated the simultaneous imaging setup to FULL array, F770W, with 8 groups and 1 integration

Proposal 11687 - Targets - EXHALE: Expelling the X-ray High-energy winds to InfrAred-traced gaLactic scale fEedback - Tracing multi...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	Mrk877-MIRI	RA: 16 20 11.2800 (245.0470000d) Dec: +17 24 27.52 (17.40764d) Equinox: J2000	Proper Motion RA: 0.02 mas/yr Proper Motion Dec: -0.020999982552893925 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Quasars, Seyfert galaxies] Extended=YES				
	(2)	Mrk877-MIRI-BACKGROUND	RA: 16 20 11.2800 (245.0470000d) Dec: +17 25 28.00 (17.42444d) Equinox: J2000	Epoch of Position: 2000	
	Comments: Category=Unidentified Description=[Blank field] Extended=YES				
	(3)	Mrk877-NIRSPEC	RA: 16 20 11.2800 (245.0470000d) Dec: +17 24 27.52 (17.40764d) Equinox: J2000	Proper Motion RA: 0.02 mas/yr Proper Motion Dec: -0.020999982552893925 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Quasars, Seyfert galaxies]				

Proposal 11687 - Observation 1 - EXHALE: Expelling the X-ray High-energy winds to InfrAred-traced gaLactic scale fEedback - Tracin...

Observation	Proposal 11687, Observation 1: Mrk877-NIRSpec											Thu Jun 11 18:56:59 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(3)	Mrk877-NIRSPEC	RA: 16 20 11.2800 (245.0470000d) Dec: +17 24 27.52 (17.40764d) Equinox: J2000				Proper Motion RA: 0.02 mas/yr Proper Motion Dec: -0.020999982552893925 mas/yr Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Quasars, Seyfert galaxies]											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		MEDIUM		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	NRSIRS2RAPID	11	3	false	true	NONE	9	27	4726.8	300586
	2	G235H/F170LP	NRSIRS2RAPID	11	3	true	false	NONE	1	3	525.2	300586
Special Requirements	Aperture PA Range 23 to 111 Degrees (V3 244.02835083 to 332.02835083) Aperture PA Range 139 to 140 Degrees (V3 0.02835083 to 1.02835083) Aperture PA Range 233 to 244 Degrees (V3 94.02835083 to 105.02835083) Aperture PA Range 338 to 349 Degrees (V3 199.02835083 to 210.02835083)											

Proposal 11687 - Observation 2 - EXHALE: Expelling the X-ray High-energy winds to InfrAred-traced gaLactic scale fEedback - Tracin...

Observation	Proposal 11687, Observation 2: Mrk877-MIRI												Thu Jun 11 18:56:59 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[Mrk877-MIRI-BACKGROUND (Obs 3)]												
Diagnostics	(Visit 2:1) Warning (Form): Data Excess over lower threshold												
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	Mrk877-MIRI	RA: 16 20 11.2800 (245.0470000d) Dec: +17 24 27.52 (17.40764d) Equinox: J2000				Proper Motion RA: 0.02 mas/yr Proper Motion Dec: -0.020999982552893925 mas/yr Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Quasars, Seyfert galaxies] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	FND	All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	8	1	1	Dither 1	4	4	88.801	300586
	1	SHORT(A)	MRSLONG		FASTR1	42	6	1	Dither 1	4	24	2852.741	300586
	1	SHORT(A)	MRSSHORT		FASTR1	42	6	1	Dither 1	4	24	2852.741	300586
	2		IMAGER	F770W	FASTR1	8	1	1	Dither 1	4	4	88.801	300586
	2	MEDIUM(B)	MRSLONG		FASTR1	42	6	1	Dither 1	4	24	2852.741	300586
	2	MEDIUM(B)	MRSSHORT		FASTR1	42	6	1	Dither 1	4	24	2852.741	300586
	3		IMAGER	F770W	FASTR1	8	1	1	Dither 1	4	4	88.801	300586
	3	LONG(C)	MRSLONG		FASTR1	42	6	1	Dither 1	4	24	2852.741	300586
	3	LONG(C)	MRSSHORT		FASTR1	42	6	1	Dither 1	4	24	2852.741	300586

Special Requirements	Sequence Observations 3, 2 (reordered), Non-interruptible
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Proposal 11687 - Observation 3 - EXHALE: Expelling the X-ray High-energy winds to InfrAred-traced gaLactic scale fEedback - Tracin...

Observation	Proposal 11687, Observation 3: Mrk877-MIRI-BACKGROUND												Thu Jun 11 18:56:59 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [Mrk877-MIRI (Obs 2)]												
Diagnostics	(Visit 3:1) Warning (Form): Data Excess over lower threshold												
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(2)	Mrk877-MIRI-BACKGROUND	RA: 16 20 11.2800 (245.0470000d) Dec: +17 25 28.00 (17.42444d) Equinox: J2000					Epoch of Position: 2000					
	Comments: Category=Unidentified Description=[Blank field] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
		All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					BACKGROUND			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	8	1	1	Dither 1	4	4	88.801	267744
	1	SHORT(A)	MRSLONG		FASTR1	42	6	1	Dither 1	4	24	2852.741	267744
	1	SHORT(A)	MRSSHORT		FASTR1	42	6	1	Dither 1	4	24	2852.741	267744
	2		IMAGER	F770W	FASTR1	8	1	1	Dither 1	4	4	88.801	267744
	2	MEDIUM(B)	MRSLONG		FASTR1	42	6	1	Dither 1	4	24	2852.741	267744
	2	MEDIUM(B)	MRSSHORT		FASTR1	42	6	1	Dither 1	4	24	2852.741	267744
	3		IMAGER	F770W	FASTR1	8	1	1	Dither 1	4	4	88.801	267744
	3	LONG(C)	MRSLONG		FASTR1	42	6	1	Dither 1	4	24	2852.741	267744
	3	LONG(C)	MRSSHORT		FASTR1	42	6	1	Dither 1	4	24	2852.741	267744

Special Requirements	Sequence Observations 3, 2 (reordered), Non-interruptible
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